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The project we chose was the marble rolling game, with rigid bodies being included within the player prefab.

Force application is in the PlayerController.cs file where we used

AddRelativeForce(desired_acceleration * impulse, 0, 0) to the player's rigid body.

We implemented hazard colliders such as the void and cacti which send the player back to their last checkpoint and checkpoint colliders which save the player's progress.

We implemented 3 physics materials: grass, sand, and ice. Grass utilizes similar physics to that of the previous assignment, while sand has higher friction and drag, and ice has little to no friction.

We implemented different trigger events such as collecting coins which increase the player's score, and an OnTriggerEnter event such that when the player reaches a checkpoint, they will respawn at that point rather than at the beginning of the game, or fall indefinitely. We also included an option to restart the game from the beginning if the player pressed a key (R).

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Darren's reflection:

My contributions were creating the collectible, creating the grass level, and creating simple assets such as the checkpoint and cactus. I also created the respawn when the player falls into the void or touches a cactus. I learned a lot about OnTriggerEvents through this project. I had primarily worked on many of the OnTrigger events and learned how to efficiently implement them. I had also learned about the importance of prefabs; creating prefabs for many of the assets saved a lot of time since all I had to do was paste them and move them to the correct position.

Alyssa:

I designed the overall level layout, built the sand and ice levels, and helped to debug various scripts in the game, such as the triggers to collect the coins to increase the score, the respawning and checkpoint systems, and the physics involved with every material. Over the course of the project, I learned how to better understand how certain functions worked within C#, and thus how to correct for any programming errors that appeared. Along the way, I also gained a better understanding of how to utilize Github for collaborative projects, of which this was the first project of this type I have worked on. Aside from level design, I also learned that not every concept will or can be reproduced in the final draft; while I originally envisioned pitfall obstacles for the player to navigate through in the sand level, this came to be not possible to create without purchasing add-ons to edit meshes within Unity, resulting in the maze found in the current iteration. I found that it was very helpful to discuss with a third individual during office hours, as explaining any problems often resulted in finding the solution either through their help or by finding the error myself while discussing where it may be coming from, both of which were great methods to better remember the information.